

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –1
TEST DATE: 20-11-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

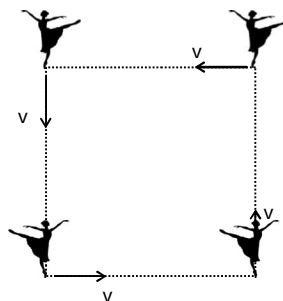
Physics

PART – I

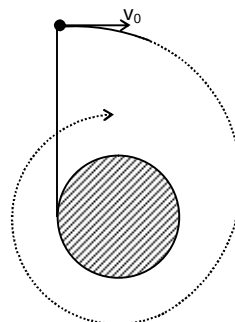
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

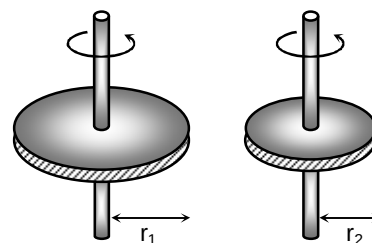
1. Four dancers start their dance by forming a square with side $\ell = 9\text{m}$. After that, they always move anticlockwise towards their neighbour with variable velocity $v = k\sqrt{x}$ where $k = 0.1\text{ m}^{1/2}\text{sec}^{-1}$ and x is the distance of the closest neighbour in meter. Then choose the correct option(s).



- (A) All the dancers will meet with each other at $t = 60\text{ sec}$ after they start their dance.
 (B) Angular velocity of the line joining the adjacent dancers increases with time.
 (C) All the dancers will meet with each other at $t = 63\text{ sec}$ after they start their dance.
 (D) Magnitude of acceleration of each dancer is remains constant with time.
2. A point object of mass $m = 2\text{kg}$ is connected to a cylinder of radius $R = 1\text{m}$ via a massless inextensible rope. At time $t = 0$ the object is moving with an initial velocity $v_0 = 5\text{ m/s}$ perpendicular to the rope as shown in the figure. The rope has initial length $L_0 = 10\text{ m}$ and having non zero tension. Figure shows the top view and the motion of the object is taking place on the horizontal smooth surface and the cylinder is always fixed. The point object moves such that the rope wraps up around the cylinder. The rope will break when the tension in the rope exceeds 100 N . Then choose the correct option(s).
- (A) Kinetic energy of the object at the instant when the rope breaks is 25 J .
 (B) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is less than $5\text{ kg-m}^2/\text{sec}$.
 (C) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is equal to $5\text{ kg-m}^2/\text{sec}$.
 (D) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is greater than to $5\text{ kg-m}^2/\text{sec}$.

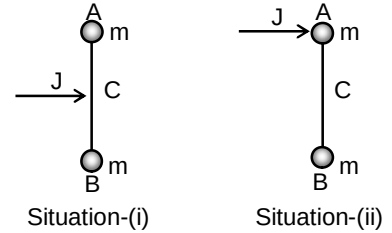


3. Two discs are mounted on thin, lightweight rods oriented through their centres and normal to the discs. These axes are constrained to be vertical at all times, and the discs can pivot frictionlessly on the rods. The discs have identical thickness and are made up of the same material, but having different radii r_1 and r_2 . The discs are given angular velocities ω_1 and ω_2 , respectively, and L_1 and L_2 represents angular momentum of the discs along their rotational axis respectively. Now they are brought into contact at the edges. After the discs interact via friction it is found that both discs come exactly to a halt. Which of the following(s) must hold? Ignore effects associated with the vertical rods.



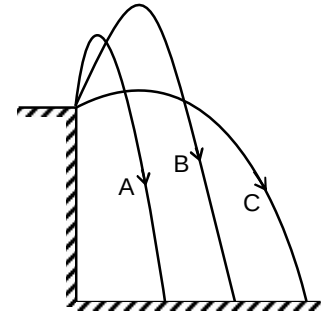
- (A) $\omega_1 r_1^4 = \omega_2 r_2^4$
 (B) $\omega_1 r_1^3 = \omega_2 r_2^3$
 (C) $\frac{L_1}{L_2} = \frac{r_1}{r_2}$
 (D) $\frac{L_1}{L_2} = \frac{\omega_1 r_1^2}{\omega_2 r_2^2}$

4. A dumbbell is formed by connecting two identical particles each of mass m with a massless rod. The system is placed on a horizontal smooth surface and impulse J is applied in two ways as shown in the figure as situation-(i) and situation-(ii). Then just after the application of impulse, choose the correct option(s).



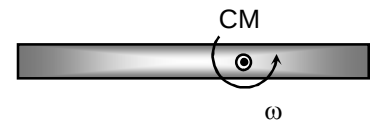
- (A) Velocity of centre of mass with respect to ground frame in situation-(i) and situation-(ii) are same.
 (B) Kinetic energy of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are same.
 (C) Angular momentum of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are different.
 (D) Velocity of particle A in situation-(ii) is double of velocity of particle A in situation-(i) just after the application of impulse.

5. Three particles A, B and C are simultaneously thrown from a cliff and their trajectories are shown in the given figure. Neglect air resistance and take gravity to be uniform during the motion. If u_A^y , u_B^y and u_C^y denotes the initial vertical components of velocity of particles A, B and C respectively and the times t_A , t_B and t_C , they take to hit the ground respectively. Then



- (A) $u_B^y > u_C^y > u_A^y$
 (B) $u_B^y > u_A^y > u_C^y$
 (C) $t_A < t_C < t_B$
 (D) $t_C < t_A < t_B$

6. A thin rod has an unknown non-uniform density. It is mounted on an axle passing perpendicular to it, through its centre of mass as shown and is then rotated about the axle. The axle divides the rod into two parts, one on each side of it. Which of the following must be true, no matter how the mass in the rod is distributed?

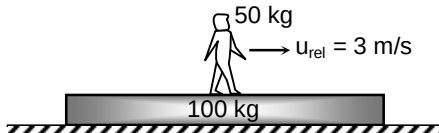
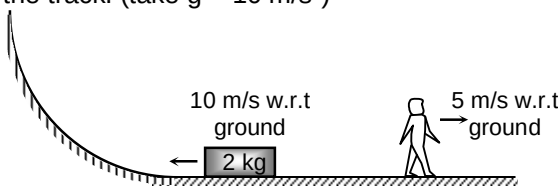
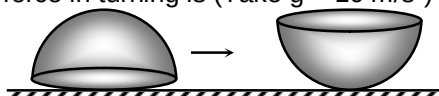


- (A) The two part must have the same mass.
 (B) The magnitudes of the linear momentum of the two parts must be equal.
 (C) The magnitudes of the angular momentum of the two parts about the centre of mass must be equal.
 (D) The kinetic energies of two parts must be equal.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

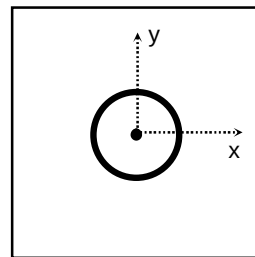
7. List –I describes for systems and List-II gives possible value of work done in Joules.

List –I		List -II	
(I)	A plank of mass 100 kg is placed on a smooth horizontal surface at rest along with a man. Now man starts moving on the plank with relative velocity of 3 m/s with respect to plank. The work done by man is 	(P)	+100J
(II)	A block of mass 2 kg is placed on a smooth track and imparted a velocity of 10 m/s to the left and observed by an observer moving with 5 m/s as shown in the figure. Work done by the normal force applied by the track on the block upto the instant when the block reaches the highest point on the track in the frame of observer. Assume that block never loose contact with the track. (take $g = 10 \text{ m/s}^2$) 	(Q)	–100J
(III)	A particle of mass 1 kg is projected from an unstable equilibrium point N towards a stable equilibrium point S with negligible velocity in a conservative force field having potential energy function $U = -\left(\frac{1}{100} \text{ J/m}^4\right)[x-10]^2[x-30]^2, \text{ where } x \text{ is in meters.}$ Then work done by conservative force during the displacement from N to S is	(R)	+150J
(IV)	A solid hemisphere of mass 4 kg and radius 10 m is slowly turned as shown in the figure. Then work done by external force in turning is (Take $g = 10 \text{ m/s}^2$) 	(S)	–150J
		(T)	Zero

Which one of the following options is correct ?

- (A) I → R, II → T, III → P, IV → P
 (B) I → S, II → T, III → Q, IV → P
 (C) I → R, II → Q, III → P, IV → P
 (D) I → T, II → Q, III → Q, IV → P

8. A disc placed on a frictionless horizontal plank is rotating with a constant angular velocity $\vec{\omega}$ about its vertical axis. Now the plank is made to move with a constant acceleration $\vec{a}$ on a straight path. Initially the centre of the disc is at the origin of xyz space which is fixed with the plank and xy plane is on the plank. List-I gives some possible $\vec{a}$, $\vec{\omega}$ and List-II gives trajectories of instantaneous centre of rotation of the disc.

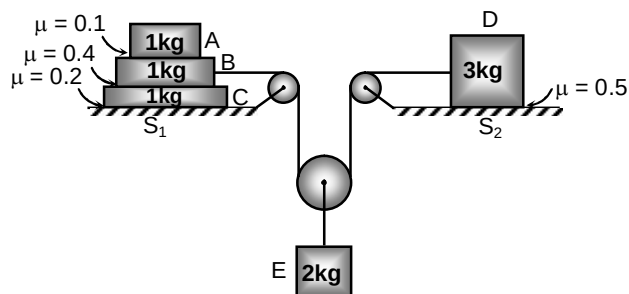


List -I		List -II	
(I)	$\vec{a} = -2\hat{j} \text{ m/s}^2, \vec{\omega} = 4\hat{k} \text{ rad/s}$	(P)	$y = 4$
(II)	$\vec{a} = -2\hat{j} \text{ m/s}^2, \vec{\omega} = -4\hat{k} \text{ rad/s}$	(Q)	$y = 4x^2 \text{ (} x \leq 0 \text{)}$
(III)	$\vec{a} = -2\hat{i} \text{ m/s}^2, \vec{\omega} = 4\hat{k} \text{ rad/s}$	(R)	$y = 4x^2 \text{ (} x \geq 0 \text{)}$
(IV)	$\vec{a} = -2\hat{i} \text{ m/s}^2, \vec{\omega} = -4\hat{k} \text{ rad/s}$	(S)	$x = 4y^2 \text{ (} y \geq 0 \text{)}$
		(T)	$x = 4y^2 \text{ (} y \leq 0 \text{)}$

Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S
 (B) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T
 (C) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ T, IV $\rightarrow$ R
 (D) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ T

9. Figure shows a system of multiple blocks placed over each other and connected through an inextensible string. All the pulleys are ideal and frictionless. List-I shows the surfaces between which the friction is asked and List-II gives the value of friction acting between the surfaces. (Take $g = 10 \text{ m/s}^2$)

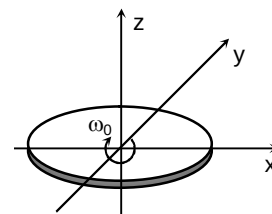


List -I		List -II	
(I)	Between A and B	(P)	15 N
(II)	Between B and C	(Q)	9.4 N
(III)	Between C and S_1	(R)	7.2 N
(IV)	Between D and S_2	(S)	6 N
		(T)	1 N

Which one of the following options is correct ?

- (A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q
 (B) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P
 (C) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ Q
 (D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ P

10. A disc of mass m is placed over a horizontal smooth surface (xy plane) such that the centre of the disc is at the origin of the xyz space, as shown in the figure. Radius of the disc is $R = 1\text{m}$ and is rotating with an angular velocity $\vec{\omega}_0 = -6\hat{k}$ rad/s and having zero velocity of centre of mass. A particle of same mass m hits the disc with a speed $|\vec{v}| = 8\text{ m/s}$ in each case of list-I. List-I gives the trajectory and velocity direction of the particle before hitting the disc and the coefficient of restitutions (e) of the collision. List-II gives the angular velocity of the disc just after the collision (in rad/s)



List -I		List -II	
(I)	$x = 0, y = R$ and $\hat{v} = -\hat{k}, e = 0$	(P)	3
(II)	$y = \frac{x}{\sqrt{3}}, z = 0$ and $\hat{v} = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}, e = 1$	(Q)	5
(III)	$y = R, z = 0$ and $\hat{v} = \hat{i}, e = 0$	(R)	6
(IV)	$y = \frac{R}{2}, z = 0$ and $\hat{v} = \hat{i}, e = 0$	(S)	7
		(T)	8

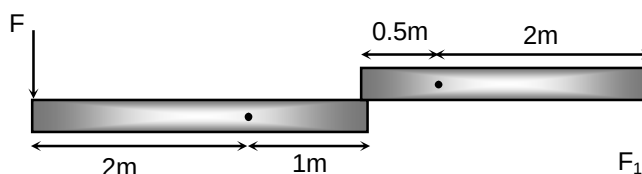
Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ T
 (B) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R
 (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q
 (D) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T

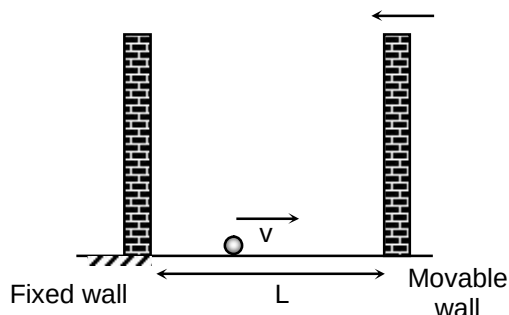
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

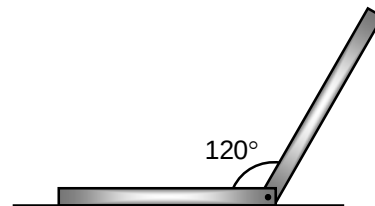
11. Two massless rods are attached to frictionless pivots, with their ends touching. The distances between the pivot points and the endpoints of the rods are shown in the figure. Neglect friction force between the rods, if a force $F = 9\text{N}$ is applied at the left end of the left rod, what force F_1 (in newton) must be applied at the right end of the right rod to keep the system in equilibrium.



12. A small sphere slides with a speed $v = 60 \text{ m/s}$ on a frictionless horizontal surface between two parallel massive walls. One of which is fixed and the other can be moved. Initial distance between walls is L . The sphere bounces between the walls perfectly elastically. Now the right wall is moved towards the left wall by a distance $0.01 L$ with speed $0.0001v$. Then what is the final speed of the sphere after the said movement of the wall in m/s .

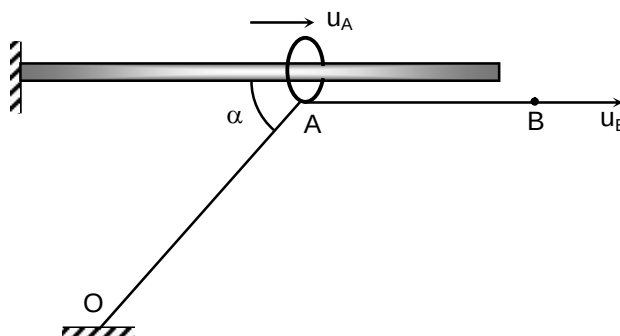


13. Figure shows a laptop made of two identical uniform plates, each of mass $m = 1 \text{ kg}$ connected by a hinge and placed on a rough horizontal surface as shown. The hinge is locked when the screen makes an angle of 120° from the keypad of the laptop. Assume that the laptop does not slide on the horizontal surface, then what is the minimum force (in newton) that can be applied on the top of the laptop, in the plane of the page to cause the bottom of the laptop to lift off the ground? (Take $g = 10 \text{ m/s}^2$)

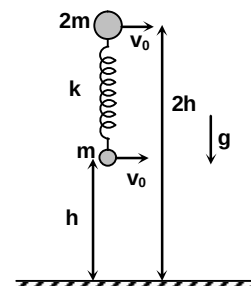


14. A block of mass $m = 1 \text{ kg}$ is dropped vertically onto a fixed ramp from a height of $h = 2.5 \text{ m}$ with angle $\theta = 37^\circ$ with horizontal and having coefficient of static friction and kinetic friction $\mu = 0.5$. The block is dropped in such a way that it does not rotate after colliding with the ramp. Through this problem assume that the time of the collision is negligible. The block slides down the ramp immediately after impact. What is the speed (in m/s) of the block right after the collision? (Take $g = 10 \text{ m/s}^2$)

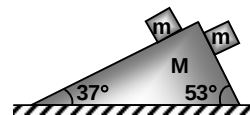
15. One end of the non stretchable rope is tied to a fixed knob O , while other end B of the rope is moving horizontally with a constant velocity $u_B = 1 \text{ m/s}$. That rope goes through a small ring A , which can move freely along a horizontal straight rod as shown in the figure. Determine velocity u_A of the ring (in m/s), when inclined section of the rope makes an angle $\alpha = 30^\circ$ with the rod.



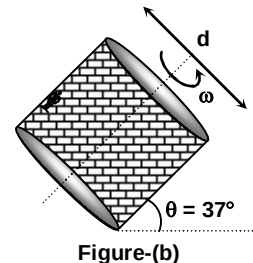
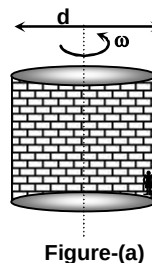
16. Two small bodies with masses m and $2m$ are attached to a spring of negligible mass. Initially the compressed spring forms a straight vertical line, when both objects are launched with same velocities $v_0 = 1 \text{ m/s}$ in the horizontal direction as shown in the figure. Initial height of the lower body is $h = 1 \text{ m}$ from the ground, while distance of heavier body is $2h = 2 \text{ m}$. After $t = 0.1 \text{ sec}$ from the launch, lower body hits the ground. Determine height H (in m) of the upper body from the ground at that moment of time. (assume no air drag and $g = 10 \text{ m/s}^2$)



17. A right triangular wooden block of mass $M = 4 \text{ kg}$ is at rest on a smooth table as shown in the figure. Two smaller wooden cubes both of mass $m = 2 \text{ kg}$, initially at rest on the two sides of the larger block start sliding down. What is the normal force (in newton) applied by the system on the table? (take $g = 9.8 \text{ m/s}^2$)



18. The 'wall of death' ride consists of a drum with diameter $d = 10\text{m}$ which rotates at an angular velocity ω as shown in the figure-(a). Passengers stand against the inside vertical rough wall and as the ride speeds up to angular velocity ω , they are held against the wall and can not slide down due to friction with the wall. The rotating drum is now tilted so that its axis of rotation is $\theta = 37^\circ$ above the horizontal as shown in the figure-(b). If the coefficient of friction between the wall and the passenger is $\mu = 0.5$, then find the minimum angular velocity ω_0 (in rad/s) of the drum so that passengers are held against the wall at the highest point. (Take $g = 10 \text{ m/s}^2$)



Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Which of the following species are polar as well as non-planar?
 (A) XeF_4
 (B) H_2O_2
 (C) $\text{XeF}_5^\ominus$
 (D) $\text{XeF}_5^\oplus$
20. All bonds are equivalent in
 (A) $\text{Cr}_2\text{O}_7^{2-}$
 (B) HCO_3^-
 (C) XeO_6^{4-}
 (D) SF_6
21. The radial part $R(r) = \frac{1}{9\sqrt{\sigma a_0^{3/2}}} \left(\frac{6r}{a_0} - \frac{r^2}{a_0^2} \right) e^{-r/3a_0}$ represent for an orbital. (a_0 is Bohr's radius of H-atom)
 Choose the correct statement about the orbital
 (A) this orbital may have one nodal plane in yz plane
 (B) it is 3p orbital
 (C) one node will be found at the $r = 6a_0$ distance from nucleus
 (D) this orbital can accommodate maximum 6 electrons
22. Which of the following geometry is most likely form from sp^3d – hybridisation ?
 (A) Linear
 (B) Tetrahedral
 (C) Bent T-shaped
 (D) See-Saw
23. Find the incorrect statement
 (A) Iodine exists as I_8^{2-} ion in Cs_2I_8
 (B) Oxidation state of Cr in CrO_5 and CrO_8^{3-} is +6
 (C) FeCl_2 is more ionic than FeCl_3
 (D) LiO_2 and NaO_2 are yellow coloured compound but KO_2 is colourless
24. Consider the following options and choose the correct one
 (A) In the given equilibrium $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
 If O_2 is added and the volume of the reaction vessel is reduced the equilibrium is shift in product side.
 (B) If we add CrO_4^{2-} ion in saturated solution of Ag_2CrO_4 it will decreases $[\text{Ag}^+]$.

- (C) Three sparingly soluble salt A_2B , AB and AB_3 are given. If all the three having same value of solubility product (K_{SP}). In the saturated solution the correct order of their solubilities is $AB > A_2B > AB_3$.
- (D) If in option (A) inert gas is added at constant pressure and temperature. Partial pressure of SO_3 will be remain same.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. If t_x represents time required for a fraction x of a reactant to be reacted during a reaction involving only one reactant. Then match List – I with List – II.

List – I		List – II	
(I)	Zero order reaction	(P)	$\frac{t_{3/4}}{t_{1/2}} = 3$
(II)	First order reaction	(Q)	$\frac{t_{1/4}}{t_{1/8}} = 2$
(III)	Second order reaction	(R)	$\frac{t_{7/8}}{t_{3/4}} = \frac{3}{2}$
(IV)	Order of reaction ≥ 1	(S)	$\alpha = (1 - e^{-kt})$
		(T)	$\frac{t_{100\%}}{t_{50\%}} = 2$

($\alpha \rightarrow$ degree of dissociation, $k \rightarrow$ rate constant)

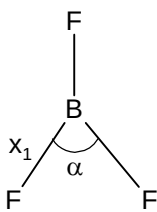
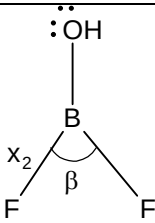
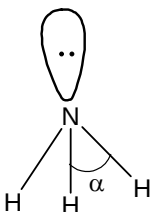
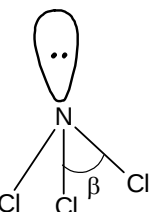
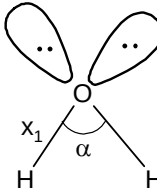
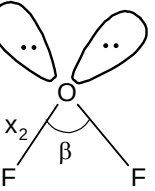
- (A) I $\rightarrow$ Q, T; II $\rightarrow$ R, S; III $\rightarrow$ P; IV $\rightarrow$ P, R, S
 (B) I $\rightarrow$ Q; II $\rightarrow$ R, T; III $\rightarrow$ P; IV $\rightarrow$ P, R
 (C) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ R, S; IV $\rightarrow$ Q, T
 (D) I $\rightarrow$ R, T; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ P, Q

26. List – I contains properties and List – II contains moles or ions

List – I		List – II	
(I)	2 lone pair on central atom	(P)	XeF_5^-
(II)	Zero dipole moment	(Q)	NF_3
(III)	Planar	(R)	ICl_3
(IV)	All adjacent bond angle are equal	(S)	XeF_4
		(T)	PCl_3F_2

- (A) I $\rightarrow$ P, R, S; II $\rightarrow$ S, T; III $\rightarrow$ P, Q, R, S; IV $\rightarrow$ P, Q, R, S, T
 (B) I $\rightarrow$ P, R, S; II $\rightarrow$ P, S, T; III $\rightarrow$ P, R, S; IV $\rightarrow$ P, Q, R, S
 (C) I $\rightarrow$ P, R, T; II $\rightarrow$ R, S, T; III $\rightarrow$ R, S; IV $\rightarrow$ S, T
 (D) I $\rightarrow$ P, S, T; II $\rightarrow$ P, R, S; III $\rightarrow$ P, Q, R, S, T; IV $\rightarrow$ P, R, S

27. List – I contains compounds with bond parameter and List – II contains bond parameter decreasing order (x_1, x_2, x_3 are bond length while α, β, γ are bond angle)

List – I			List – II	
(I)			(P)	$x_1 > x_3 > x_2$
(II)			(Q)	$\gamma > \alpha > \beta$
(III)			(R)	$\alpha > \beta > \gamma$
(IV)	(S – S bond length) S_8 S_2F_2 S_2Cl_2 x_1 x_2 x_3		(S)	$x_3 > x_2 > x_1$
			(T)	$\gamma > \beta > \alpha$

- (A) I $\rightarrow$ S, T; II $\rightarrow$ R; III $\rightarrow$ Q, S; IV $\rightarrow$ S
 (B) I $\rightarrow$ R, S; II $\rightarrow$ Q; III $\rightarrow$ S, T; IV $\rightarrow$ P
 (C) I $\rightarrow$ R, S; II $\rightarrow$ R; III $\rightarrow$ Q, S; IV $\rightarrow$ P
 (D) I $\rightarrow$ R, S; II $\rightarrow$ T; III $\rightarrow$ S, T; IV $\rightarrow$ S

28. List – I contains elements and compounds in respective properties and List – II contains decreasing order of properties.

List – I					List – II	
(I)	B	Al	Ga	In	(P)	$4 > 3 > 2 > 1$
	1	2	3	4		(Atomic size)
(II)	NaOH	KOH	RbOH	CsOH	(Q)	$4 > 2 > 3 > 1$
	1	2	3	4		(Thermal stability)
(III)	SiO ₂	CO ₂	CaO	Cs ₂ O	(R)	$1 > 2 > 3 > 4$
	1	2	3	4		(Basic strength)
(IV)	BeF ₂	BaF ₂	CaF ₂	MgF ₂	(S)	$4 > 3 > 1 > 2$
	1	2	3	4		(Water solubility)
					(T)	$2 > 3 > 4 > 1$

- (A) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T
 (B) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ T
 (C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R
 (D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

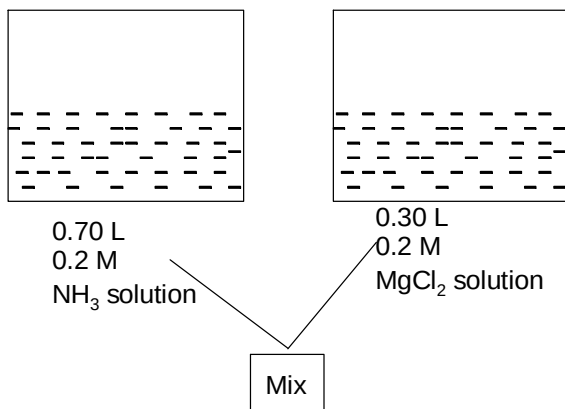
29. Fauzosite, a zeolite has formula $\text{Na}_x \left[(\text{AlO}_2)_y (\text{SiO}_2)_{136} \right] \cdot 250\text{H}_2\text{O}$
The value of $\frac{x}{y}$ is....
30. Calculate the pH at equilibrium point when a solution of 0.6 M CH_3COOH is titrated with a solution of 0.20 M NaOH (K_a of CH_3COOH 10^{-5} at 25°C ($\log 2 = 0.30$, $\log 3 = 0.48$))
31. In cyanuric triazide $[\text{C}_3\text{N}_3(\text{N}_3)_3]$. The number of sp^2 hybridised centre atoms is/are 'x' and number of π bond that can lie in same plane is/are 'y'. Calculate the value of $\frac{x}{y}$?
32. For reaction $\text{I}^- + \text{ClO}_3^- + \text{H}_2\text{SO}_4 \longrightarrow \text{Cl}^- + \text{HSO}_4^- + \text{I}_2$
If stoichiometric coefficient of HSO_4^- is 'x' and total number of electron gain in the balance reduction half reaction is 'y'. Calculate the value of $\frac{x+z}{y}$?
(z – Stoichiometric coefficient of I_2).
33. 2 mole of $\text{NH}_4\text{HS}(\text{s})$ is taken in close container of volume 2 litre. If 1 mole $\text{NH}_4\text{HS}(\text{s})$ taken part in reaction to attain the equilibrium state and at equilibrium moles of $\text{H}_2(\text{g})$ is 0.75.

$$\text{NH}_4\text{HS}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g}) \quad K_{C_1} = x$$

$$\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g}) \quad K_{C_2} = y$$

 At equilibrium $[\text{NH}_3] = a \text{ M}$
 $[\text{N}_2] = b \text{ M}$ calculate the value of $\frac{y_1}{x_1(a+b)}$ is...?
34. The mechanism of the reaction $\text{X} + 2\text{Y} \xrightarrow{K, E_a, A} \text{P}$ is
 Step 1. $\text{X} + \text{Y} \xrightleftharpoons[K_2, E_{a_2}, A_2]{K_1, E_{a_1}, A_1} \text{I}$ (Fast)
 Step 2. $\text{I} + \text{Y} \xrightarrow{K_3, E_{a_3}, A_3} \text{P}$ (Slow)
 If the value of $K_1, K_2, K_3, E_{a_1}, E_{a_2}, E_{a_3}, A_1, A_2$ and A_3 are given below
- | K_1 | K_2 | K_3 | A_1 | A_2 | A_3 | E_{a_1} | E_{a_2} | E_{a_3} |
|-----------|-----------|-----------|-----------------|-----------------|-----------------|-----------|-----------|-----------|
| 10^{-3} | 10^{-4} | 10^{-5} | 2×10^8 | 2×10^9 | 2×10^6 | 20 | 40 | 35 |
- Rate constant unit and pre-exponential factors for the reaction is in sec^{-1} and activation energy is in kJ. Calculate the value of $\frac{K \times A}{E_a}$ is....?

35.



What mass (in g) of (NH₄)₂SO₄ should be added to the above mixture cause the Mg(OH)₂ to re-dissolve? (K_{SP} for Mg(OH)₂ = 6.0×10^{-12} and K_b for NH₃ = 2×10^{-5})

36.

In a elementary chemical reaction $A(aq) \xrightarrow{t_{1/2}=10 \text{ min}} 2B(aq) + C(aq)$

Reaction mixture react with another reactant T and n-factor of A, B and C react with T in the ratio 2 : 1 : 2 respectively.

If volume of 'T' used at 10 min is 40 ml. Calculated volume of T used at $t = 0$ is x ml, at $t = 20$ min, is y ml at $t = \infty$ is z ml.

Find out the value of $\frac{y+z}{x}$ is.....?

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. If maximum value of $f(x) = \frac{3\sqrt{2x^2 + x}}{2x^2 + x + 2\sin^2 \theta}$ (where $x > 0$) is $\frac{3}{\sqrt{2}}$, then θ may lie in
- (A) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$ (B) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
 (C) $\left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$ (D) $(-\pi, \pi)$
38. If $\int_0^x f(t) dt = e^x - ae^{2x} \int_0^1 f(t) e^{-t} dt$, then
- (A) $a = \frac{1}{3 - 2e}$ (B) $f(x) = e^x - 2e^{2x}$
 (C) $a = \frac{1}{e}$ (D) $f(x) = e^x - e^{-x}$
39. If a function satisfies $(x - y)f(x + y) - (x + y)f(x - y) = 2(x^2y - y^3) \forall x, y \in \mathbb{R}$ and $f(1) = 2$, then
- (A) $|f(x)|$ is non differentiable at two points (B) $|f(x)|$ is continuous everywhere
 (C) $f(-4) = 12$ (D) $f(4) = 18$
40. The value(s) 'a' for which the area of the triangle included between the axes and any tangent to the curve $x^a y = \lambda^a$ is constant is/are
- (A) $-\frac{1}{2}$ (B) -1
 (C) $\frac{1}{2}$ (D) 1
41. Let $f(x)$ be non-positive continuous function and $F(x) = \int_0^x f(t) dt \forall x \geq 0$ and $f(x) \geq c F(x)$ where $c > 0$ and let $g : [0, \infty) \rightarrow \mathbb{R}$ be a function such that $\frac{d(g(x))}{dx} < g(x) \forall x > 0$ and $g(0) = 0$, then
- (A) the total number of roots of the equation $f(x) = g(x)$ is 1
 (B) the total number of roots of the equation $f(x) = g(x)$ is 0
 (C) the number of solution of equation $|x^2 + x - 6| = f(x) + g(x)$ is 1
 (D) the number of solution of equation $|x^2 + x - 6| = f(x) + g(x)$ is 0
42. If $y(\sin 2x + y \cos x) + \sin x \frac{dy}{dx}(\sin x + 2y) = \sin x(1 + 2\cos x)$ and $\left(\frac{\pi}{2}, 0\right)$ is a point on the curve, then which of the following is a point on the curve also
- (A) $\left(\frac{3\pi}{2}, -1\right)$ (B) $\left(\frac{3\pi}{2}, 0\right)$
 (C) $\left(\frac{\pi}{2}, -1\right)$ (D) none of these

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match the following List-I with List-II

List-I		List-II	
(I)	If function f satisfies $\int_0^1 (f^4(x) - 8x^2 f^2(x)) dx + \frac{16}{5} = 0$, then the value of $f^2(1)$ is	(P)	0
(II)	If the point of inflection for the curve $y = \int_0^x \sqrt{1-t^4} dt$ is (α, β) , then $\alpha + \beta$ is equal to	(Q)	1
(III)	Let f be continuous and function g is defined as $g(x) = \int_0^x \left(t^2 \cdot \int_1^x f(u) du \right) dt$ also $f(1) = 3$, and $g'(1) + g''(1)$ is equal to $n_1 + \frac{f'(1)}{n_2}$ (where $n_1, n_2 \in \mathbb{N}$), then $n_1 - n_2$ is	(R)	2
(IV)	Let $I_1 = \int_0^1 \frac{e^x}{1+x} dx$ and $I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)} dx$, then $\frac{I_1}{eI_2}$ is	(S)	3
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

44. Match the following List-I with List-II

List-I		List-II	
(I)	Number of points of discontinuity of $f(x) = \left[\frac{2x}{\pi} \right] + \{\sin x\}$ in $x \in [0, \pi]$ is/are (where $[.]$ and $\{.\}$ represents greatest integer function and fractional part function)	(P)	0
(II)	If the area included between the curve $y^2 = x^3$, tangent to it at point (1, 1) and x-axis is A, then $[A + 1]$ (where $[.]$ represents greatest integer function)	(Q)	1
(III)	If the area bounded by $f : [1, \infty) \rightarrow [1, \infty)$ given by $f(x) = x^2 - 2x + 2$ and its inverse is A, then $3A$ is equal to	(R)	2
(IV)	Let $I_n = \int_0^1 x^n \sqrt{1-x^2} dx$, then $\lim_{n \rightarrow \infty} \frac{I_n}{I_{n-2}}$ is equal to	(S)	3
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (Q)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

45. Match the following List-I with List-II

List-I		List-II	
(I)	Let $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$, then the value of $\int_{1/4}^{3/4} f(f(x))dx$ is	(P)	5
(II)	Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^n$, $n \in \mathbb{N}$. If area between $f(x)$ and $g(x)$ in first quadrant is $\frac{1}{2}$, then n is	(Q)	$\frac{11}{2}$
(III)	If $f(x) = \frac{1}{1+e^x}$ and $S = f(5) + f(4) + f(3) + \dots + f(-3) + f(-4) + f(-5)$, then the value of S is	(R)	$\frac{13}{2}$
(IV)	If $f(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{\frac{1}{7}}$, then $f(f(4))$	(S)	$\frac{1}{4}$
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (Q)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (S)

46. Match the following List-I with List-II

List-I		List-II	
(I)	If $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function such that $f(x + 2y) = f(x) + f(2y) + 4xy$ $\forall x, y \in \mathbb{R}$, then $f'(1) - f'(0)$ is equal to	(P)	1
(II)	If $\lim_{x \rightarrow \infty} f(x)$ exist and is finite and non-zero and $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$, then value of $\lim_{x \rightarrow \infty} f(x)$ is	(Q)	2
(III)	Let $f(x) = 1 + 2\cos x + 3\sin x$. If real number a, b, c are such that $af(x) + bf(x+c) = 1$ holds for any $x \in \mathbb{R}$, then $\left \frac{bc \cos c}{a} \right $ is	(R)	3
(IV)	Define $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x+y) = 3^y f(x) + 2^x f(y) \forall x, y \in \mathbb{R}$ and $f(1) = 1$, then $f(2)$ is	(S)	4
		(T)	5

The correct option is:

- (A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (P)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. Function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is differentiable and satisfies the condition $f(x) - f(y) \geq \ln\left(\frac{x}{y}\right) + x - y$, for all $x, y \in \mathbb{R}^+$. If g denote the derivative of f , then the value of the $g\left(\frac{1}{2022}\right)$
48. Consider a function $f(x) = \prod_{r=1}^{\infty} \left(\frac{1 + 2\cos\left(\frac{2x}{3^r}\right)}{3} \right)$. Then two number of points where $|x f(x)| + ||x - 2| - 1|$ is non differentiable on interval $(0, 3\pi)$
49. If the value of definite integral $I = \int_0^2 (3x^2 - 3x + 1)\cos(x^3 - 3x^2 + 4x - 2)dx$ can be expressed in the form of $\alpha(\sin \beta)$, where $\alpha, \beta \in \mathbb{N}$, then the value of $\frac{\alpha^2 + \beta^2 + 1}{2}$
50. If $S = \sum_{r=1}^{502} \left\lfloor \frac{305r}{503} \right\rfloor$, then $\frac{S}{10,000}$ is (where $[.]$ represents greatest integer function)
51. The value of $I = \frac{2^{2022} \int_0^1 x^{1010} (1-x)^{1010} dx}{\int_0^1 x^{1010} (1-x^{2022})^{1010} dx}$ is
52. Let f be a differentiable function satisfying the functional equation $f(xy) = f(x) + f(y) + \frac{x+y-1}{xy}$
 $\forall x, y > 0$ and $f'(1) = 2$, then the value of $[f(e^{200})]$ is equal to (where $[.]$ represents greatest integer function)
53. Consider the following area:
 A_1 = Area of region common to $y \leq f(x)$ and $y \geq g(x)$
 A_2 = Area of region common to $y \geq f(x)$ and $y \leq g(x)$
 A = Area of region bounded by $y = f(x)$, $y = g(x)$ and $x = \pm\pi$
 where $f(x) = |\cos |x||$, $g(x) = |\sin |x||$ for $x \in [-\pi, \pi]$. Then the value of $(\sqrt{2} + 1)(A_1 + A_2 + A_3)$ is
54. If $I = \int_0^{\pi/2} \sin x \log_e (\sin x) dx$ and A is area of the region bounded by the set $\left\{ (x, y) : 0 \leq y \leq \min\left\{3 - x, x + \frac{1}{x}\right\} \right\}$, then the value of $A - I$ is